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RESEARCH ARTICLE

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Comparative evaluation of medullation types and their respective fiber diameter in alpacas from seven regions of Peru

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ABSTRACT

Alpaca fibers may be classified according to the type of medullation, into fibers with strongly medullated (StrMed), continuous (ContMed), discontinuous (DiscMed), fragmented (FragMed), and fibers without medulla (NoMed). ContMed and StrMed may cause prickling and issues during dyeing and are less resistant in textile processing; therefore, they should be eliminated from fleece. Thus, to produce more comfortable alpaca fibers, it is essential to establish a baseline, and for this reason, this research aimed to evaluate the incidence of fibers according to medullation type and their respective average fiber diameters (AFD) in Peruvian regions. Samples of alpaca fibers were collected from Apurímac, Arequipa, Cusco, Huancavelica, Junín, Pasco and Puno, which were analyzed using the Fiber Med equipment of the Textile Fiber Laboratory of the Natural Fiber's Tech SAC, a Technological Research and Development Center. Medullation and fiber diameter data did not have a normal distribution, then for statistical analysis, nonparametric techniques were applied. The means of the percentages of medullation and fiber diameters were: 61.87% and 19.12 μm; 16.84% and 22.62 μm; 7.56% and 24.87 μm; 13.23% and 28.56 μm; and 0.50 and 45. 69 μm for NoMed, FragMed, DiscMed, ContMed, and StrMed, respectively; showing greater variation in medullation types than medullated fiber diameter. The geographical region, owner type, and age of animals influenced the medullation types and their fiber diameters ($p < 0.05$). On the other hand, ContMed was similar by breed; the sex only influenced FragMed. AFD_FragMed and AFD_StrMed were similar by breed. AFD_StrMed was also similar by sex. The percentages of medullation and the fiber diameters of alpacas from the seven Peruvian regions maintain high variability by location, age, owner type, breed and sex. Furthermore, there is a positive correlation between the percentage of medullation and its fiber diameters, with ContMed and StrMed fibers showing greater diameters.

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Alpaca; breeding; fiber quality; medullation; objectionable fibers

Introduction

The cuticle, cortex and medulla are the structural components of an animal fiber (Czaplicki, 2012; Markova, 2019; Tridico 2009; Villarroel, 1963). Depending on this structure, the fiber quality varies and depends on the fiber diameter, standard deviation of the fiber diameter, and medullation mainly (Quispe et al., 2013). The presence of medulla is very frequent in alpaca fibers, and runs along the internal axis; fibers with fragmented, discontinuous, continuous medulla can be found and can occupy more than 60% of the fiber diameter, calling the latter strongly medullated fibers (Balasingam & Mahar, 2005; Pinares et al., 2018). Logically, some fibers do not have a medulla and are those with a smaller average diameter (Pinares et al., 2019; Quispe et al., 2023) compared to medullated fibers.

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The medulla has particular characteristics regarding thermal insulation and specific density. The thermal insulation of the fibers is important for animals that live in cold conditions (Moore, 2015; Wang et al., 2005), because they do not require additional energy to maintain thermal homeostasis. Garments made with alpaca fiber also maintain the wearer's thermal insulation, keeping them comfortable in both cold and hot climates. The density of wool is influenced by the degree of medullation: 1.307 g/cm^3 for non-medullated wool, 1.241 g/cm^3 for wool with a low degree of medullation, and 1.035 g/cm^3 for highly medullated wool (Merrick & Scobie, 1997). Thus, the lower specific density of alpaca fibers compared to wool (1.309 versus 1.320 g/cm^3) may be related to their higher proportion of medullated fibers (Czaplicki, 2012). This gives garments made from alpaca fiber a lightness and good drape.

However, strongly medullated fibers and some with continuous medullation are generally considered undesirable or objectionable, since they appear as contaminants in the fabric (Sánchez et al., 2016) and would be responsible for producing itching when they are in contact with the skin (Cruz et al., 2024), also they are less resistant and lead to a lower performance in textile processing (Gupta et al., 1981), and additionally, they affect the transmission of light through the fiber, resulting in a heterogeneous and uneven mottled appearance in fabrics after dyeing, reducing the value of the garments when they contain this type of fibers (Hunter, 1993). Samples of alpaca fibers without strongly medullated fibers have been found, but some that would have up to 16% of strongly medullated fibers, while the percentage of fibers with continuous medullation would vary between 0.73 and 69%, with averages between 10.46 and 25.79% (Cruz et al., 2024; Quispe et al., 2023; Olarte et al., 2023).

In alpacas from the Huancavelica and Puno regions, fibers with fragmented medullation would have a percentage between 9.52 and 18.16%, while fibers with discontinuous medullation would be between 7.60 and 16.34%, respectively; additionally, the fineness of these fibers would be very close to that of non-medullated fibers, and therefore, the fibers that would have an average diameter greater than 30 µm (comfort factor limit) would be very reduced, therefore, these types of fiber would be considered desirable as well as non-medullated fibers (Cruz et al., 2024; Quispe et al., 2023). In alpacas raised in a herd in Poland, fiber medullation was found to be 60% and discontinuous fibers had an incidence of 40%. It was observed that in light-colored fibers the incidence of medullation was lower than in dark-colored fibers (Radzik-Rant & Wiercińska, 2021), and that fibers with a higher degree of medullation would be found in the neck area (Radzik-Rant et al., 2021); however, Aruquipa (2015), working with samples of colored fibers in alpacas from Bolivia, found that the incidence of medullation did not exceed 30%.

There is currently a strong consensus on selecting animals to eliminate and/or reduce the incidence of continuous medullation and strongly medullated fibers (Cruz et al., 2024; Pinares et al., 2018; Quispe et al., 2023), with the aim to produce alpaca fiber that is comfortable against the skin when garments made from it are worn .Therefore, a study with a larger number of alpacas is required, to serve as a baseline for the incidence of fibers according to their medullation type.

In alpacas, few works have been carried out on the percentages of fibers according to their medullation type, mainly due to the laboriousness required in its determination when using the projection microscope (Quispe et al., 2023), and although the OFDA 100 allowed the percentage of medullation and mohair and wool fibers to be evaluated quickly, it was later found that it did not have good accuracy when evaluating alpaca fibers (Pinares et al., 2018; Quispe et al., 2023). In many cases, there are publications on this subject considering small sample sizes, or with few fiber counts per sample. In other cases, information for a specific geographic area in Peru or from other countries is considered, however before the work reported here there has been a lack of data from the main alpaca-producing regions in Perú.

For these reasons, research was carried out on the type of medullation and diameter of fibers from seven regions of Peru (Apurímac, Arequipa, Cusco, Huancavelica, Junín, Pasco and Puno). Factors considered included the type of owner, sex, age, category, breed and geographic region. Different types of medullation and the fiber diameter were measured and correlations determined.

Material and methods

Location and duration

Alpaca fiber samples were collected from November to December 2022, from the seven regions of Peru: Apurímac, Arequipa, Cusco, Huancavelica, Junín, Pasco and Puno, which are considered the main producers of this South American camelid; while the fiber evaluations were carried out in the Textile Fiber Laboratory of the Natural Fiber's Tech SAC, located

in Lima, Peru, beginning in March, culminating in September 2023.

Animal source and sampling

The alpacas were born between January and March of each year, and the shearing was done between November and December. The fiber samples (between 7 and 10 g of weight) were taken during shearing, from the mid-side site, centered over the 10th rib midway between the back line and belly line (Aylan-Parker & McGregor, 2002). Then, each sample was stored in a paper bag, recording the sex (male and female), and age category according to the type of dentition (4 categories have been considered), where A = MT (milk teeth, 7 to 18 months), $B = 2$ T (two teeth, 18 months to 2 years), $C = 4$ T (four teeth, 2–3 years), and D = FM (full mouth, ≥3 years), region (seven geographic regions of Peru), owner type [Communal, Agricultural Society of Social Interest (ASSI), and University] and breed (Huacaya and Suri), as specified in Table 1.

Sample analysis

Before evaluation, the samples were washed, dried and conditioned at 20 ± 2 °C and 65 ± 3% relative humidity. The samples were then prepared according to the procedure indicated by IWTO-8-04 (IWTO, 2017), for which fiber fragments (between 0.2 and 0.4 mm in length) were obtained using a Hardy microtome and a thin blade. These fragments were carefully placed into the middle of a slide, delimited by an accessory called a 'fiber placer', and dispersed with slow circular motions using a glass rod and a drop of immersion oil (with a refractive index very similar to that of the fiber). Finally, a coverslip was placed on the fiber fragments to prevent the formation of air bubbles (Quispe et al., 2020).

The measurement of the average diameter and medullation of fiber sample was carried out with an electronic and intelligent device, called Smart Fiber Med, according to the procedure indicated by Quispe et al. (2020) and Quispe et al. (2023). Fiber Med evaluated an average of 1667 fibers per sample in 40 s, with medullation being classified as: non-medullated (NoMed); fragmented medullation (FragMed); discontinuous medullation (DiscMed); continuous medullation (ContMed); strongly medullated (StrMed); and total medullated (TMed = FragMed + DiscMed + ContMed + StrMed). NoMed, FragMed, DiscMed, ContMed, StrMed and TMed fibers, expressed in absolute and relative units (quantity: 'n', and percentage: '%', respectively). Additionally, it obtains the global average fiber diameter (AFD) and by medullation category (AFD_NoMed, AFD_FragMed, AFD_DiscMed, AFD_ContMed, AFD_StrMed).

Statistical analysis

An exploratory analysis of the percentages of fibers with different types of medullation and their respective fiber diameters was performed to detect outliers and confirm normal distribution. The normality of the data was verified using the Shapiro-Wilk test ($\alpha = 0.05$). Because the data were not normally distributed, the Kruskal-Wallis test was used as a non-parametric alternative to make comparisons between independent groups (Kruskal & Wallis, 1952). When differences between groups were found, multiple

Table 1. Number of samples collected from white alpacas, taking into account the region, type of owner, age category, breed and sex of the animal.

	Region							
Factor, levels	Apurímac	Arequipa	Cusco	Huancavelica	Junín	Pasco	Puno	Total
Type of owner								
Communal	450	533	192	–	–	197	–	1372
ASSI	–	–	–	–	266	–	394	660
University	–	–	81	153	–	–	–	234
Age category								
Milk teeth	146	252	101	7	–	58	–	564
Two teeth	108	90	58	32	44	35	–	367
Four teeth	90	143	42	41	5	24	–	345
Full mouth	106	48	72	73	217	80	394	990
Breed								
Huacaya	410	531	239	153	266	197	194	1990
Suri	40	2	34	–	–	–	200	276
Sex								
Female	228	426	183	153	197	158	197	1542
Male	222	107	90	–	69	39	197	724
Total	450	533	273	153	266	197	394	2266

ASSI: Agricultural Society of Social Interest; -: Data not available.

comparison were performed using the Dunn test ($\alpha = 0.05$). The Spearman's correlation coefficients were also obtained between the types of medullation and their respective diameters ($\alpha = 0.05$). All these analyses were performed with the free software R. v 4.3.2 (R Core Team, 2024).

Results and discussion

General description

The mean of percentage TMed and AFD were 38.13% and 21.33 µm, respectively, showing more variation in medullation incidence (CV = 63.73%) than AFD (CV = 17.91%) (Table 2). The percentage of fibers with different medullation types showed means of 61.87, 16.84, 7.56, 13.23 and 0.50% for NoMed, FragMed, DiscMed, ContMed, and StrMed, respectively; the AFDs found were of 19.12, 22.62, 24.87, 28.56, and 45.69 µm for AFD_NoMed, AFD_FragMed, AFD_DiscMed, AFD_ContMed, and AFD_StrMed, respectively, showing that as the fiber medulla becomes more noticeable, the AFD increases. Strongly medullated fibers are the thickest, followed by ContMed, DiscMed, and fibers with fragmented medulla are the thinnest (Table 2); this is in agreement with the results of Paucar et al. (2024) for white alpaca fibers, they also evidenced the same pattern for the medulla diameters.

Additionally, it was found that 30% the alpacas have less than 21% of TMed, and 68% of the alpacas have less than 50% of TMed, respectively, but 50% of alpacas have medullation percentages between 15.5 and 55.5%. Likewise, 23% of animals do not have StrMed, while 51% have 0.1% or less of StrMed (Figure 1). The TMed values are similar to those reported by McGregor (2006) for alpacas from Australia, who found that nearly one-third of Huacaya and Suri saddles had less than 20% of their fibers medullated. However, 68% of our alpacas had less than 50% TMed, whereas McGregor (2006) found that only 30% of alpacas had less than 50% TMed. Regarding fineness, we observed that alpacas in Peru have finer fibers than those in Australia. McGregor (2006) found that about 50% of alpaca fleece had an AFD greater than 29.9 µm. However, we found that only 50% of alpaca fleeces had an AFD greater than 26.3 µm. Pinares et al. (2019) reported higher percentages of ContMed, DiscMed, and FragMed in male alpacas from Pacomarca in Puno, although the AFDs for these fibers were similar to our results.

The percentage of TMed, NoMed and ContMed fibers showed higher variation than other type of medullated fibers. Likewise, the AFDs by type of medullation had high variability for StrMed fibers, the ranges are very wide, from 24.21 to 79.89 µm. Radzik-Rant and Wiercińska (2021), working with alpacas from Poland, reported percentages of medullation and AFD higher to the findings in this study. On the other hand, Pinares et al. (2018) reported a higher percentage of TMed (67.4%) by projection microscopy in male Huacaya alpacas from Pacomarca in Peru, while Min et al. (2020) showed AFD of 24.15 µm with 44.7% of medullation in fiber from alpacas in China. Therefore, our findings demonstrate better quality of alpaca fiber in Peru. Apart from the genetic factor (Pinares et al., 2018), feeding has little influence on the percentage of medullation (Lupton et al., 1991). Medullation is formed in the fetal stage before nutritional or environmental factors influence unless the possibility of malnutrition occurs at the follicular level (Wilson, 1929). The percentages and AFD of different fiber types according their medullation were similar than those found by Quispe et al. (2023), but to percentage of StrMed were lower in this study (1.3 versus 0.5%). In addition, the percentages of different fiber

Table 2. Descriptive statistics of the different types of medullation and their respective fiber diameters in alpacas.

Variables	*N*	Average	Median	Standard deviation	Minimum	Maximum	Asymmetry	Kurtosiss
TMed (%)	2266	38.13	33.72	24.30	0.54	100.00	0.50	−0.76
SE TMed (%)	2266	1.06	1.06	0.35	0.00	4.30	0.86	5.82
NoMed (%)	2266	61.87	66.29	24.30	0.00	99.46	−0.50	−0.76
FragMed (%)	2266	16.84	16.05	9.63	0.00	57.58	0.41	−0.47
DiscMed (%)	2266	7.56	5.42	6.78	0.00	49.13	1.47	2.46
ContMed (%)	2266	13.23	9.33	12.62	0.00	82.70	1.87	4.02
StrMed (%)	2266	0.50	0.10	1.11	0.00	10.61	4.10	21.00
AFD (µm)	2266	21.33	20.89	3.82	12.54	36.58	0.58	0.36
SD AFD (µm)	2266	5.00	4.80	1.24	1.73	11.73	1.15	2.16
AFD_NoMed (µm)	2265	19.12	18.86	3.02	12.00	32.74	0.59	0.50
AFD_FragMed (µm)	2264	22.62	22.29	2.53	15.72	37.23	0.94	1.95
AFD_DiscMed (µm)	2264	24.87	24.55	2.63	13.59	45.75	1.28	5.57
AFD_ContMed (µm)	2264	28.56	28.09	3.15	19.55	47.89	1.01	2.20
AFD_StrMed (µm)	1736	45.69	45.83	9.96	24.21	79.89	0.22	−0.05

TMed: Total medullation; SE: Standard error; NoMed: non-medullated; FragMed: fragmented medullation; DiscMed: discontinuous medullation; ContMed: continuous medullation; StrMed: Strongly medullated; AFD: Average fiber diameter; SD: standard deviation.

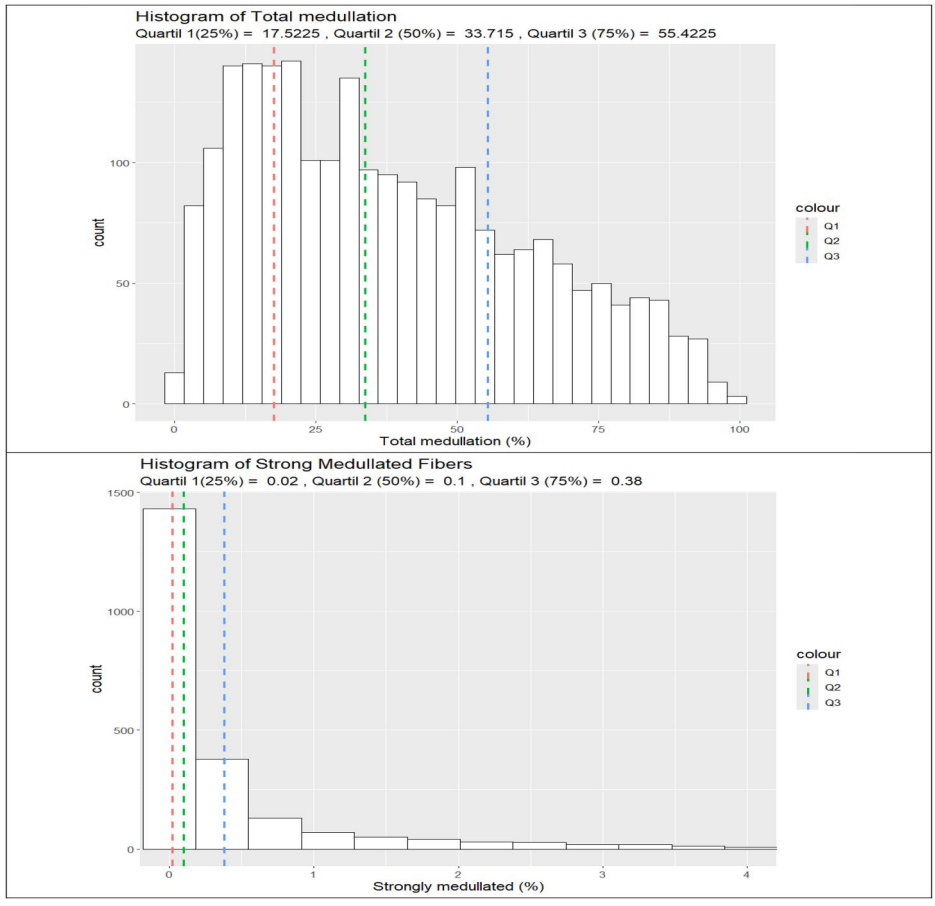

Figure 1. Graphics of total medullation and strongly medullated fibers with quartiles 1, 2 and 3.

types according their medullation were similar than those found by Quispe et al. (2023) using projection microscope and Smart Fiber Med, but also similar than found by Ormachea et al. (2024) and Pinares et al. (2024) who worked in alpacas from Puno and Cusco, respectively using Smart Fiber Med.

Cruz et al. (2019) found values of 37.32% and 26.60 µm for the percentage of TMed and AFD, respectively, in Huacaya alpacas from Pacomarca. The data on the percentage of TMed are similar than ours, but the AFD is higher in the study by Cruz et al. (2019). Lupton et al. (2006) found an AFD of 27.85 µm for Huacaya alpacas from the United States and the TMed was 18.48%, therefore, our results of TMed are higher with lowers AFD. This difference could be from the OFDA 100 instrument used to calculate the medullation. Otherwise, the incidence of different types of medullation and their respective fiber diameters found in our study are similar to the reports of Cruz et al. (2024) in Huacaya alpacas from Pacomarca. Radzik-Rant and Wiercińska (2021) worked with Huacaya alpacas from Poland and reported higher total medullation and fiber diameter values that we did, 68.91% with 25.21 µm. In addition, the percentage of fibers with different types of medullation and their respective diameters vary between studies due to differences in environment, diet, genetic constitution, and management.

The ContMed and StrMed fibers together represent about 14% and are those with the thickest AFDs, 28.56 and 45.69 µm, respectively. In particular, these diameters have a wide range of variation between 19.55 and 47.89 µm for ContMed fibers and between 24.21 and 79.89 µm for StrMed fibers. A significant portion of these fibers would be those that cause itching because the thickest fiber diameters protrude from the garment. The prickling is a negative aspect, which is why the industry prefers fine alpaca fibers that do not cause prickle. Cruz et al. (2019) state that alpaca fiber could be more attractive to the textile industry if the prickling factor were reduced. A more sustainable alternative to removing objectionable fibers would be to select against the percentage of StrMed and ContMed fibers, which were the most prevalent fibers (57.5% of fibers with ContMed and 36.8% of StrMed fibers) in objectionable alpaca fibers (Paucar et al., 2024), thus improving the comfort and

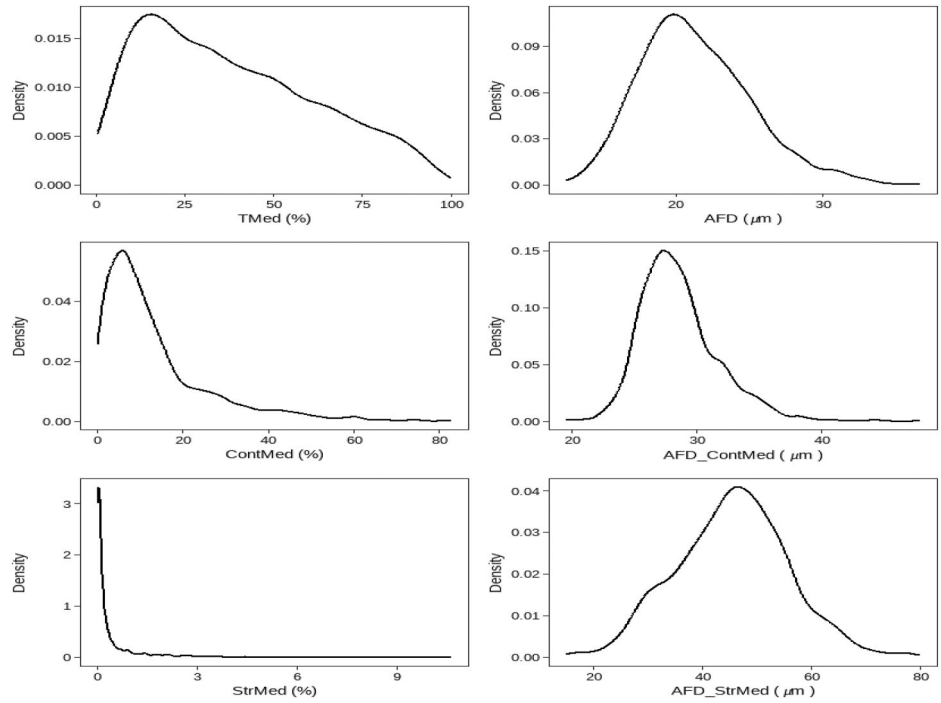

Figure 2. Density curves of medullation percentages and fiber diameters of alpaca. TMed: total medullation; ContMed: continuous medullation; StrMed: strongly medullated; AFD: average fiber diameter; AFD_ContMed: AFD of continuous medullated; AFD_StrMed: AFD of strongly medullated.

attractiveness of alpaca fiber in the national and international market.

On the other hand, the density curves for medullation percentage and fiber diameters did not show a symmetrical distribution (Figure 2), this finding is consistent with the results of Apaza and Quispe (2020), who reported a gamma distribution for the fiber diameter of alpacas from Puno; therefore, the commonly used parametric statistical methods would not be suitable for variables related to fiber quality. Most of the variables show a positive asymmetrical distribution, that is, there are high values, both for the percentages of fibers by type of medullation, as well as for their fiber diameters; but they present low frequencies; except for the percentage of NoMed fibers, which presents a negative asymmetry (−0.50), which indicates that it has a tail to the left.

Influence of the factors on the medullation and fiber diameter

The percentage of TMed, NoMed, FragMed, DiscMed, ContMed and StrMed fibers were significantly affected ($p < 0.05$) by the region, type of owner and age category (Table 3). Quispe et al. (2023) reported the influence of location, breed and age on the TMed, but Pinares et al. (2024) reported influence of location, age and sex on TMed, AFD and different types of fiber according their medullation, with exception of age and sex that do not have effect on ContMed and StrMed, While other authors reported the influence of various factors, such as Min et al. (2020) demonstrated the influence of location on the percentage of medullated fibers, Berolatti et al. (2021) reported the influence of age on the percentage of medullation. Breed also showed effects on the different types of medullation ($p < 0.05$), except for the percentage of ContMed fibers ($p > 0.05$). The morphological difference in the fleece between Huacaya and Suri is known, but at the microscopic level, the cause and why the Huacaya fiber is more medullated than that of Suri is unknown (Villarroel, 1963). At the genomic level, 16 genomic regions associated with differences in the type of fleece in Huacaya and Suri alpacas were identified in the chromosome. These findings suggest an important influence of keratin on the physical and textile structure of alpaca fiber (Tan et al., 2024).

However, sex did not influence the percentages of medullation studied ($p > 0.05$), except the percentage of FragMed fibers ($p < 0.05$). Quispe et al. (2023) also found no effect of sex on the percentage of TMed. Lupton et al. (2006) demonstrated that age and region caused medullation to vary, but no differences were

Table 3. Means ± SE (Standard Error) and medians (in bold) of the percentage of fibers for different types of medullation, and their fiber diameters by region, breeding level, age category, breed and sex of alpaca.

Factor/level	TMed	NoMed	FragMed	DiscMed	ContMed	StrMed
Medullation (%)						
Region	[<0.001]	[<0.001]	[<0.001]	[<0.001]	[<0.001]	[<0.001]
Apurímac	36.6 ± 0.94 **31.9b**	63.4 ± 0.94 **68.1c**	16.52 ± 0.41 **14.96bc**	6.72 ± 0.24 **5.07 cd**	13.03 ± 0.51 **9.54 cd**	0.32 ± 0.04 **0.08c**
Arequipa	26.1 ± 0.92 **20.15**[a]	73.9 ± 0.92 **79.85d**	13.55 ± 0.44 **10.66**[a]	5.1 ± 0.25 **2.99**[a]	7.32 ± 0.34 **5.23a**	0.14 ± 0.01 **0.04b**
Cusco	33.67 ± 1.23 **31.5b**	66.33 ± 1.23 **68.5c**	15.15 ± 0.51 **14.6b**	6.24 ± 0.32 **4.82bc**	11.73 ± 0.62 **8.9c**	0.55 ± 0.07 **0.11d**
Huancavelica	44.3 ± 1.83 **43.2c**	55.7 ± 1.83 **56.8b**	17.59 ± 0.58 **17.05c**	7.79 ± 0.49 **6.72de**	18.39 ± 1.14 **14.58ef**	0.53 ± 0.07 **0.2a**
Junín	58.14 ± 1.34 **59.78d**	41.86 ± 1.34 **40.22**[a]	22.81 ± 0.51 **22.25d**	12.39 ± 0.44 **11.21f**	21.87 ± 0.98 **16.06f**	1.07 ± 0.09 **0.45e**
Pasco	35.77 ± 1.78 **30.4b**	64.23 ± 1.78 **69.6c**	19.46 ± 0.83 **18.8c**	6.38 ± 0.48 **3.93b**	9.78 ± 0.73 **6.62b**	0.15 ± 0.02 **0.04b**
Puno	44.5 ± 1.3 **43.15c**	55.5 ± 1.3 **56.85b**	17.2 ± 0.45 **17.05c**	9.98 ± 0.41 **7.54e**	16.37 ± 0.72 **12.49de**	0.96 ± 0.08 **0.27a**
Type of owner	[<0.001]	[<0.001]	[<0.001]	[<0.001]	[<0.001]	[<0.001]
Communal	31.96 ± 0.59 **27.71**[a]	68.04 ± 0.59 **72.29c**	15.61 ± 0.27 **13.94**[a]	5.97 ± 0.15 **4**[a]	10.12 ± 0.27 **7.26a**	0.26 ± 0.02 **0.06a**
ASSI	49.99 ± 0.98 **49.68c**	50.01 ± 0.98 **50.32**[a]	19.46 ± 0.36 **19.56c**	10.95 ± 0.31 **9.07c**	18.58 ± 0.59 **14.19b**	1.00 ± 0.06 **0.33c**
University	40.83 ± 1.43 **39.94b**	59.17 ± 1.43 **60.06b**	16.66 ± 0.47 **16.41b**	7.29 ± 0.37 **6.01b**	16.36 ± 0.85 **12.86b**	0.52 ± 0.06 **0.19b**
Age category	[<0.001]	[<0.001]	[<0.001]	[<0.001]	[<0.001]	[<0.001]
Milk teeth	21.31 ± 0.72 **15.99**[a]	78.69 ± 0.72 **84.01c**	9.82 ± 0.3 **7.47**[a]	3.27 ± 0.14 **2.31**[a]	8.04 ± 0.38 **5.65a**	0.18 ± 0.02 **0.05a**
Two teeth	37.1 ± 1.1 **33.9b**	62.9 ± 1.1 **66.1b**	17.52 ± 0.46 **16.89b**	6.79 ± 0.29 **4.9b**	12.27 ± 0.57 **9.79b**	0.52 ± 0.07 **0.09c**
Four teeth	35.91 ± 1.11 **31.01b**	64.09 ± 1.11 **68.99b**	17.58 ± 0.49 **16.11b**	6.86 ± 0.29 **4.86b**	11.13 ± 0.53 **7.45b**	0.34 ± 0.04 **0.06b**
Full teeth	48.86 ± 0.77 **48.35c**	51.14 ± 0.77 **51.65**[a]	20.33 ± 0.29 **20.12c**	10.52 ± 0.24 **8.71c**	17.27 ± 0.46 **12.49c**	0.74 ± 0.04 **0.21d**
Breed	[0.024]	[0.024]	[<0.001]	[0.002]	[0.093]	[<0.001]
Huacaya	38.58 ± 0.55 **34.66b**	61.42 ± 0.55 **65.34**[a]	17.42 ± 0.22 **16.96b**	7.73 ± 0.15 **5.6b**	13.05 ± 0.28 **9.23**	0.38 ± 0.02 **0.08a**
Suri	34.86 ± 1.37 **30.4**[a]	65.14 ± 1.37 **69.6b**	12.63 ± 0.43 **11.86**[a]	6.32 ± 0.35 **4.55**[a]	14.53 ± 0.83 **11.08**	1.37 ± 0.12 **0.52b**
Sex	[0.943]	[0.943]	[0.017]	[0.881]	[0.772]	[0.363]
Female	37.95 ± 0.6 **34.27**	62.05 ± 0.6 **65.73**	17.16 ± 0.25 **16.46b**	7.52 ± 0.17 **5.51**	12.82 ± 0.3 **9.31**	0.45 ± 0.02 **0.1**
Male	38.51 ± 0.95 **32.2**	61.49 ± 0.95 **67.8**	16.16 ± 0.36 **14.99**[a]	7.64 ± 0.26 **5.08**	14.09 ± 0.54 **9.35**	0.61 ± 0.05 **0.1**
Fiber diameter (μm)						
Region	[<0.001]	[<0.001]	[<0.001]	[<0.001]	[<0.001]	[<0.001]
Apurímac	19.68 ± 0.13 **19.36**[b]	17.29 ± 0.09 **17.25**[a]	21.22 ± 0.08 **21.45**[a]	23.92 ± 0.06 **24.01**[a]	27.24 ± 0.09 **27.03**[a]	42.52 ± 0.56 **43.7**[a]
Arequipa	18.76 ± 0.13 **18.52**[a]	17.08 ± 0.1 **16.96**[a]	21.72 ± 0.08 **21.63**[b]	24.62 ± 0.1 **24.35**[b]	28.59 ± 0.14 **27.9**[b]	44.5 ± 0.58 **43.53**[a]
Cusco	21.72 ± 0.16 **21.47**[c]	19.71 ± 0.12 **19.64**[bc]	23 ± 0.11 **22.82**[d]	25.49 ± 0.11 **25.27**[c]	29.1 ± 0.13 **28.95**[c]	47.43 ± 0.59 **46.49**[bc]
Huancavelica	24.01 ± 0.27 **23.74**[d]	21.34 ± 0.21 **21.38**[d]	24.73 ± 0.19 **24.62**[e]	26.81 ± 0.2 **26.59**[d]	30.64 ± 0.24 **30.16**[d]	45.21 ± 0.68 **45.62**[ab]
Junín	26.48 ± 0.21 **25.93**[e]	23.29 ± 0.16 **22.98**[e]	25.71 ± 0.18 **25.48**[e]	27.51 ± 0.21 **27.63**[d]	31.55 ± 0.22 **31.59**[d]	49.01 ± 0.63 **50.42**[c]
Pasco	21.47 ± 0.21 **20.97**[c]	20.05 ± 0.16 **19.65**[c]	22.66 ± 0.15 **22.29**[cd]	24.16 ± 0.2 **23.91**[a]	27.56 ± 0.22 **27.28**[a]	50.53 ± 1.11 **49.89**[c]
Puno	21.81 ± 0.15 **21.71**[c]	19.42 ± 0.11 **19.22**[b]	22.24 ± 0.11 **22.09**[c]	23.65 ± 0.12 **23.64**[a]	27.34 ± 0.14 **27.24**[a]	44.21 ± 0.54 **44.48**[a]
Type of owner	[<0.001]	[<0.001]	[<0.001]	[<0.001]	[<0.001]	[0.003]
Communal	19.83 ± 0.08 **19.62**[a]	17.91 ± 0.07 **17.81**[a]	21.84 ± 0.05 **21.8**[a]	24.43 ± 0.06 **24.28**[a]	28.05 ± 0.07 **27.63**[a]	44.88 ± 0.35 **44.96**[a]
ASSI	23.69 ± 0.15 **23.4**[b]	20.98 ± 0.12 **20.47**[b]	23.65 ± 0.12 **23.2**[b]	25.21 ± 0.13 **24.79**[b]	29.04 ± 0.14 **28.75**[b]	46.38 ± 0.42 **47.12**[b]
University	23.42 ± 0.21 **23.18**[b]	20.98 ± 0.16 **20.72**[b]	24.29 ± 0.15 **24.06**[c]	26.45 ± 0.15 **26.04**[c]	30.21 ± 0.19 **29.86**[c]	45.58 ± 0.57 **45.61**[ab]
Age category	[<0.001]	[<0.001]	[<0.001]	[<0.001]	[<0.001]	[0.004]
Milk teeth	18.57 ± 0.11 **18.38**[a]	17.14 ± 0.09 **17.09**[a]	21.51 ± 0.08 **21.61**[a]	24.27 ± 0.1 **24.22**[a]	28.18 ± 0.13 **27.64**[a]	44.41 ± 0.54 **44.58**[a]
Two teeth	21.07 ± 0.17 **20.67**[c]	18.81 ± 0.14 **18.61**[c]	22.5 ± 0.12 **22.19**[c]	24.8 ± 0.11 **24.55**[b]	28.39 ± 0.14 **27.84**[a]	45.18 ± 0.63 **44.98**[ab]
Four teeth	20.55 ± 0.18 **20.18**[b]	18.38 ± 0.15 **18.2**[b]	22.15 ± 0.13 **21.85**[b]	24.77 ± 0.12 **24.27**[b]	28.07 ± 0.16 **27.49**[a]	45.03 ± 0.64 **45.52**[ab]
Full teeth	23.26 ± 0.11 **22.78**[d]	20.63 ± 0.09 **20.17**[d]	23.46 ± 0.09 **22.97**[d]	25.26 ± 0.1 **24.87**[c]	29.02 ± 0.11 **28.74**[b]	46.16 ± 0.36 **46.99**[b]
Breed	[0.018]	[<0.001]	[0.679]	[<0.001]	[0.009]	[0.995]
Huacaya	21.28 ± 0.09 **20.84**[a]	19.06 ± 0.07 **18.77**[a]	22.63 ± 0.06 **22.28**	24.98 ± 0.06 **24.62**[b]	28.64 ± 0.07 **28.12**[b]	45.41 ± 0.28 **45.83**
Suri	21.66 ± 0.19 **21.44**[b]	19.56 ± 0.15 **19.55**[b]	22.55 ± 0.13 **22.35**	24.02 ± 0.14 **23.77**[a]	27.99 ± 0.16 **27.62**[a]	45.55 ± 0.48 **45.46**
Sex	[0.022]	[0.001]	[<0.001]	[0.007]	[0.011]	[0.476]
Female	21.37 ± 0.09 **21.05**[b]	19.23 ± 0.08 **19**[b]	22.72 ± 0.06 **22.41**[b]	24.91 ± 0.07 **24.66**[b]	28.63 ± 0.08 **28.21**[b]	45.35 ± 0.3 **45.81**
Male	21.24 ± 0.16 **20.54**[a]	18.89 ± 0.11 **18.55**[a]	22.41 ± 0.1 **22.07**[a]	24.77 ± 0.1 **24.28**[a]	28.41 ± 0.12 **27.79**[a]	45.61 ± 0.45 **45.71**

The *p*-values of the Kruskal-Wallis test for each factor are shown in brackets. Different letters in the columns for each factor indicate significant differences by the Dunn test. ($p < 0.05$). TMed: Total medullation; *SE*: standard error; NoMed: non-medullated; FragMed: fragmented medullation; DiscMed: discontinuous medullation; ContMed: continuous medullation; StrMed: Strongly medullated. ASSI: Agricultural Society of Social Interest.

found by sex and color in alpacas from the United States. On the other hand, Lozano et al. (2023) reported higher percentages of TMed in black and brown alpacas compared to white. It is understood that the percentage of fibers by type of medullation varies, depending on the area studied. Therefore, they should be taken into account in a possible genetic improvement program that involves the measurement of medullation percentages, in its different types, in addition to their respective fiber diameters.

The percentage of total medullation was higher in alpacas from the Junín region ($p < 0.05$) compared to other regions; however, alpacas from the Arequipa region showed a lower percentage of medullation ($p < 0.05$), followed by Cusco and Apurimac regions, then Pasco, Puno and Huancavelica. This same pattern is shown for the different types of medullation; but an opposite pattern is shown for the percentage of NoMed fibers, this was to be expected since it is the complement of the percentage of TMed.

Regarding the type of owner, alpacas from ASSI show higher percentages of TMed ($p < 0.05$), followed by Universities and Communities, with significant differences between them for the percentage of TMed, FragMed, DiscMed and StrMed fibers. Regarding age, alpacas with full mouth showed a higher percentage

of TMed, NoMed, FragMed, DiscMed and ContMed fibers ($p < 0.05$), followed by two and four teeth animals (without differences between them), and milk teeth animals maintain low percentages of medullation. For the percentage of StrMed fibers, full teeth animals present a higher proportion ($p < 0.05$), followed by two teeth, four teeth and milk teeth animals, with significant differences between them. These results are congruent with those obtained by Ormachea et al. (2024) and Pinares et al. (2024). McGregor (2006) showed linear increases of 10 to 60% in medullation with fleece weight and fiber diameter, also increasing from 20 to 36 µm in fiber from both Huacaya and Suri alpaca from Australia.

Suri alpacas had a lower percentage of TMed, FragMed and DiscMed fibers ($p < 0.05$); but they presented a higher proportion of StrMed fibers ($p < 0.05$), around four times the percentage of StrMed fibers of Huacaya alpacas. These results are consistent with those of Quispe et al. (2023) and Pinares et al. (2024), evidencing the differentiation between Huacaya and Suri alpacas for this characteristic. Therefore, this should be taken into account in possible genetic improvement programs that aim to reduce the prickling factor, based on the reduction of the proportion of fibers with different types of medullation. The percentage of fibers with different types of medullation for females and males was similar ($p > 0.05$), except for the percentage of FragMed fibers, which were higher in females; nevertheless Radzik-Rant and Wiercińska (2021) reported a higher percentage of TMed in males.

The global AFD and AFD by fiber medullation types were influenced ($p < 0.05$) by the geographic region of Peru, type of owner, and age category (Table 3). Min et al. (2020) found an influence of location on fiber diameter. Breed significantly influenced fiber diameters with different medullation types ($p < 0.05$), except for the diameter of FragMed and StrMed fibers ($p > 0.05$). Sex is a factor that also had a significant influence ($p < 0.05$) on AFDs, except for the AFD_StrMed fibers ($p > 0.05$). Pinares et al. (2023) reported an influence of breed, place of origin, and age on global AFD; but they found no influence of sex, a result very similar to our findings. Lupton et al. (2006) reported significant effects of sex, age, region, and color on global AFD in alpacas from the United States. Paucar-Chanca et al. (2019) found an influence of age on mean fiber diameter; but found no influence of sex in alpacas from the National University of Huancavelica.

The global AFD is higher in animals from the Junín region ($p < 0.05$) compared to the other regions, while the finest fibers were found in alpacas from Arequipa, Apurímac, Cusco, Pasco and Puno ($p < 0.05$); alpacas from Huancavelica have global AFD thicker than other regions, but finer than alpacas from Junín. The AFD by medullation type also differed between regions ($p < 0.05$), with alpacas from Huancavelica and Junín having the AFD thicker than alpacas of another region; except for AFD_StrMed, where Pasco and Junín were thicker. McGregor (2006) reported the influence of the year on this characteristic in alpacas from Australia. Paucar-Chanca et al. (2019) found a global AFD of 23.25 µm, which is consistent with the results of this study for alpacas managed by the national universities. Radzik-Rant and Wiercińska (2021) found thicker fibers in males than in females in alpacas from Poland.

According to owner type, the finer fibers were found in alpacas raised in peasant communities than in ASSI and university alpacas ($p < 0.05$). Also, the AFD_FragMed, AFD_DiscMed and AFD_ContMed were lower in the peasant communities, followed by the ASSI and universities ($p < 0.05$). On the other hand, the AFD_StrMed fibers of the peasant communities were lower than the ASSI only ($p < 0.05$), but similar those of the Universities. Although in the ASSI and Universities, productive and reproductive records are used, the animals are subjected to better management and animal selection is carried out, globally the fiber diameter has not yet been reduced, compared to alpacas managed at the communal level.

Young animals showed a smaller fiber diameter for each type of medullation ($p < 0.05$). As the age of the animal increases, more shearing is also performed, which causes an increase in fiber diameter. On the other hand, the global AFD of Suri alpacas was found to be higher than Huacaya ($p < 0.05$), which is consistent with the reported by Pinares et al. (2023). However, the AFD_DiscMed and AFD_ContMed were higher in Huacaya than in Suri ($p < 0.05$), although no differences were found for the AFD_FragMed and AFD_StrMed according to breed ($p > 0.05$). Likewise, there were no differences in AFD_StrMed by sex ($p > 0.05$), but male alpacas showed smaller global AFD, AFD_FragMed, NoMed, AFD_FragMed, AFD_DiscMed and AFD_ContMed than females ($p < 0.05$).

Correlations

The Spearman's correlations between the percentage of TMed, percentages of the different types of

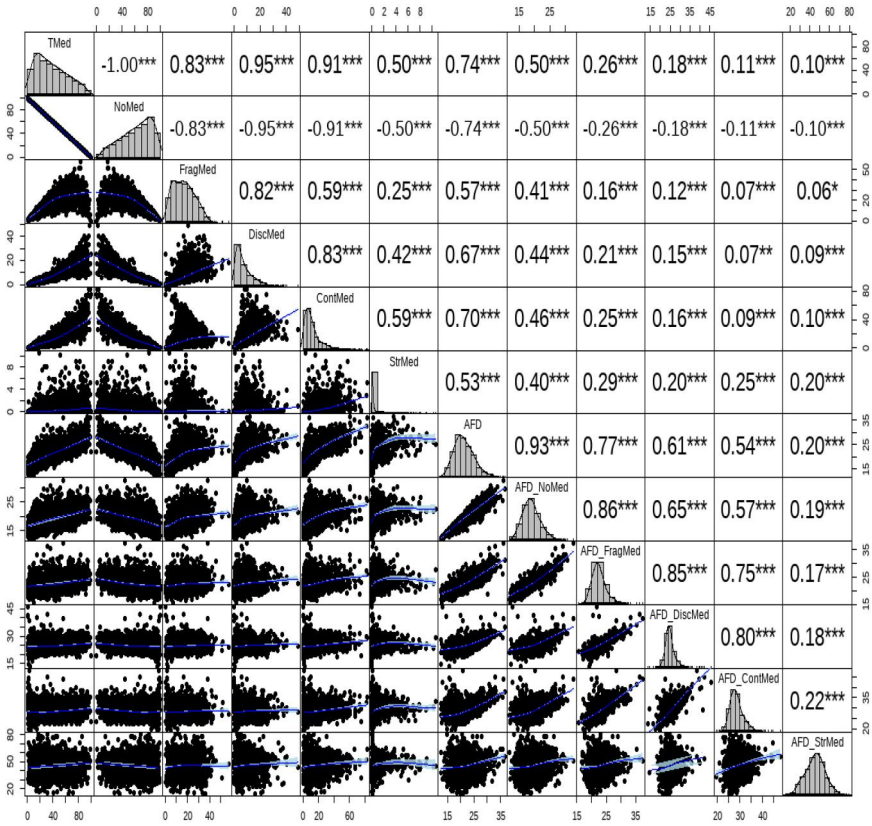

Figure 3. Spearman's correlations of medullation percentages and their respective fiber diameters. TMed: Total medullation; NoMed: Non-medullated; FragMed: fragmented medullation; DiscMed: discontinuous medullation; ContMed: continuous medullation; StrMed: Strongly medullated; AFD: average fiber diameter. ***Indicate significant correlations ($p < 0.01$).

medullation, global AFD and AFD by type of medullation showed significant values ($p < 0.05$) in all cases (Figure 3). These correlations were positive, except the correlations between the percentage of NoMed fibers with the other variables, which were negative correlations. The correlations between the global AFD with the percentage of TMed, FragMed, DiscMed, ContMed and StrMed were high, 0.74, 0.57, 0.67, 0.70 and 0.53, respectively. Berolatti et al. (2021) found a direct association of the global AFD with the percentages of fibers with different types of medullas for alpaca, paco-vicuña and angora rabbit. Also, Pinares et al. (2019) working with male alpacas from Pacomarca-Puno, found a high correlation (0.80) between the global AFD and the percentage of TMed, this value is slightly higher than our results (0.74). Cruz et al. (2019) found genetic correlations of 0.53 and 0.55 between the percentage of TMed and global AFD in Suri and Huacaya alpacas, respectively.

Cruz et al. (2024) reported positive correlations of global AFD with AFD by type of medullation, similar to our results; however, they found negative associations between the percentage of StrMed fibers with percentages of FragMed, DiscMed and ContMed. Also, they found a negative correlation between the percentage of FragMed with ContMed (−0.78), different from our result (0.59), in which all correlations were positive. These wide differences would indicate that the correlations could be associated with the different study populations; in any case, more research is necessary in this regard.

To make alpaca fiber more attractive for the textile industry, it is necessary to reduce the prickling factor and medullated fibers (Pinares et al., 2019; Cruz et al., 2024). StrMed and many ContMed fibers are known as objectionable fibers (Quispe and Quispe, 2024); these are found in wool, mohair, cashmere, alpaca and llama, and they constitute a contamination problem for the textile industry (Sánchez et al., 2016; Hunter et al., 2013), because they increase the prickle sensation and heterogeneity of fabrics, reducing their quality. In Poland, Radzik-Rant and Wiercińska (2021) proposed that breeding alpacas with fine and non-medullated fibers should be the focus of breeders; while Pinares et al. (2018) suggest selection against medullated fibers, because it would indirectly reduce fiber diameter, and Min et al. (2020) suggest improving the content of non-medullated fibers and would

be the goal for refining the fiber. However, the evidence from this study shows that fibers with continuous medulla and strongly medullated are those with the largest diameter and would be those that contribute greatly to the prickling factor; in addition, Paucar et al. (2024) have shown that the highest percentage of objectionable fibers are occupied by fibers with continuous medullation and strongly medullated. Therefore, it is necessary to take into account fibers with both types of medullation (ContMed and StrMed fibers) in an eventual selection program.

Berolatti et al. (2021) suggest that the percentage of TMed fibers should be reduced in llamas and alpacas, and should be considered as a selection criterion in genetic improvement programs. To reduce the percentage of ContMed and StrMed fibers, the measurement of this parameter should be mandatory and continuously monitored, otherwise it would be impossible to produce exclusive quality fibers in alpacas (Quispe et al., 2023). The implementation of the monitoring of alpaca fiber medullation is still a pending task; for which the joint action of state institutions, non-governmental organizations, public universities and producers is necessary.

Conclusions

The medullation percentages and average fiber diameters of alpacas in the seven main regions of Peru maintain high variability, and they are affected by location, age category, type of owner, breed, and sex. Direct relationships are evident between the percentage of fibers with different types of medullation and their corresponding average fiber diameters. The presence of fibers with higher diameters that are continuously and strongly medullated suggests changing the selection criteria against these fibers in an alpaca breeding program.

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